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(Q1) Write a program TO add TWO numbers 05h &07h ?

ORG 0000H ; start of program

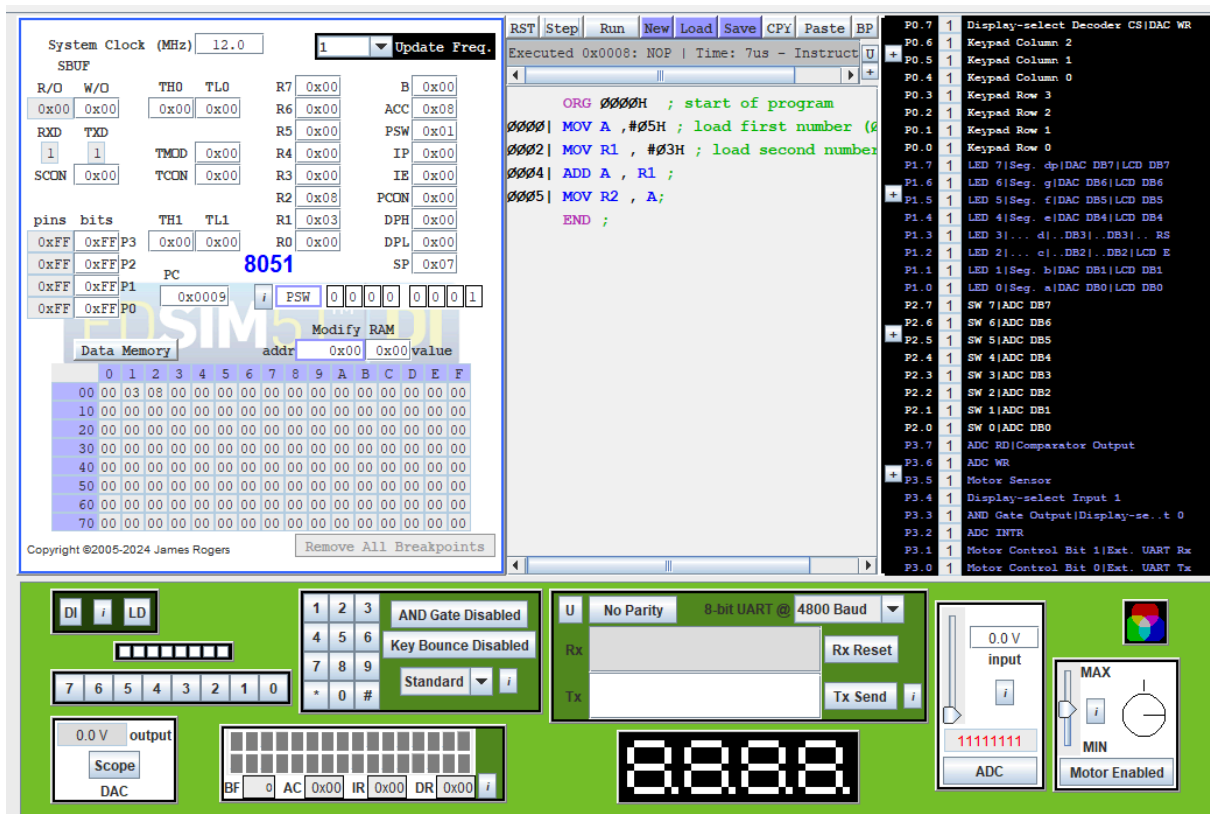
0000| MOV A ,#05H ; load first number (05H)

0002| MOV R1 , #03H ; load second number (03H)

0004| ADD A , R1 ;

0005| MOV R2 , A;

END ;



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(Q2) WRITE A PROGRAM TO SUBTRACT TWO NUMBER 05H AND 07H ?

ORG 0000H ;

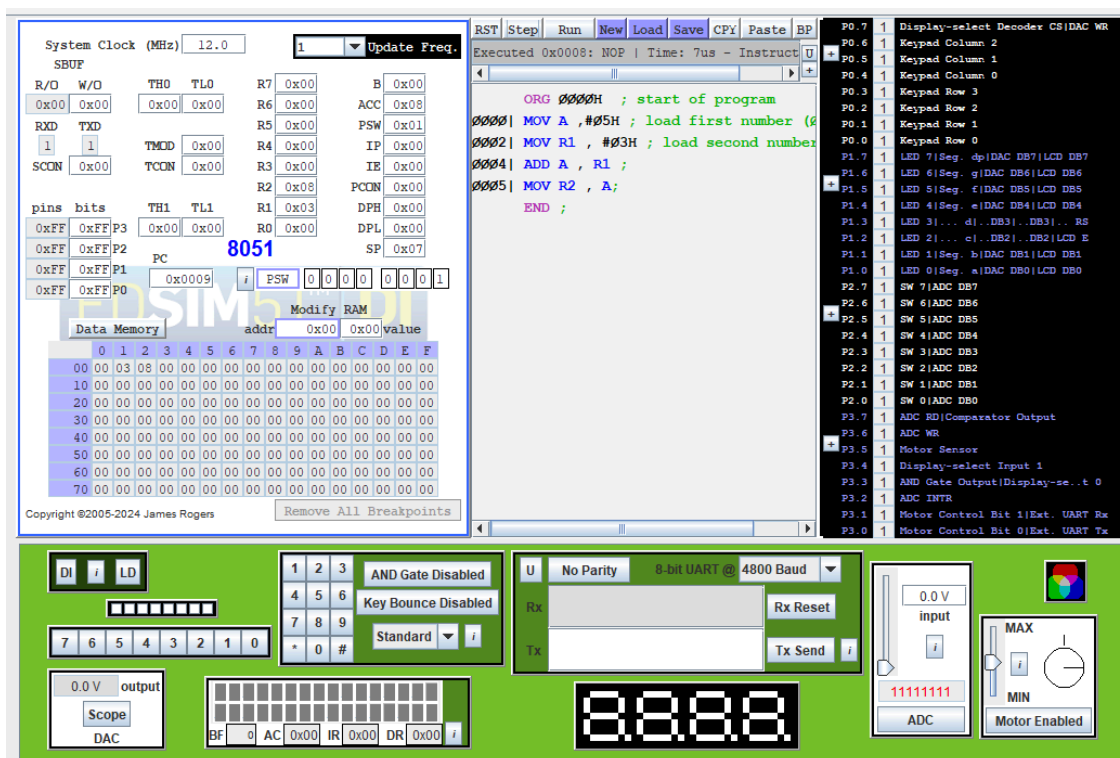
0000| MOV A, #07H ;

0002| MOV R1, #05H ;

0004| SUBB A, R1 ;

0005| MOV R2, A ;

END ;



(Q3) WRITE A POROGRAM TO MULTIPLY TWO NUMBER 05H AND 08H ?

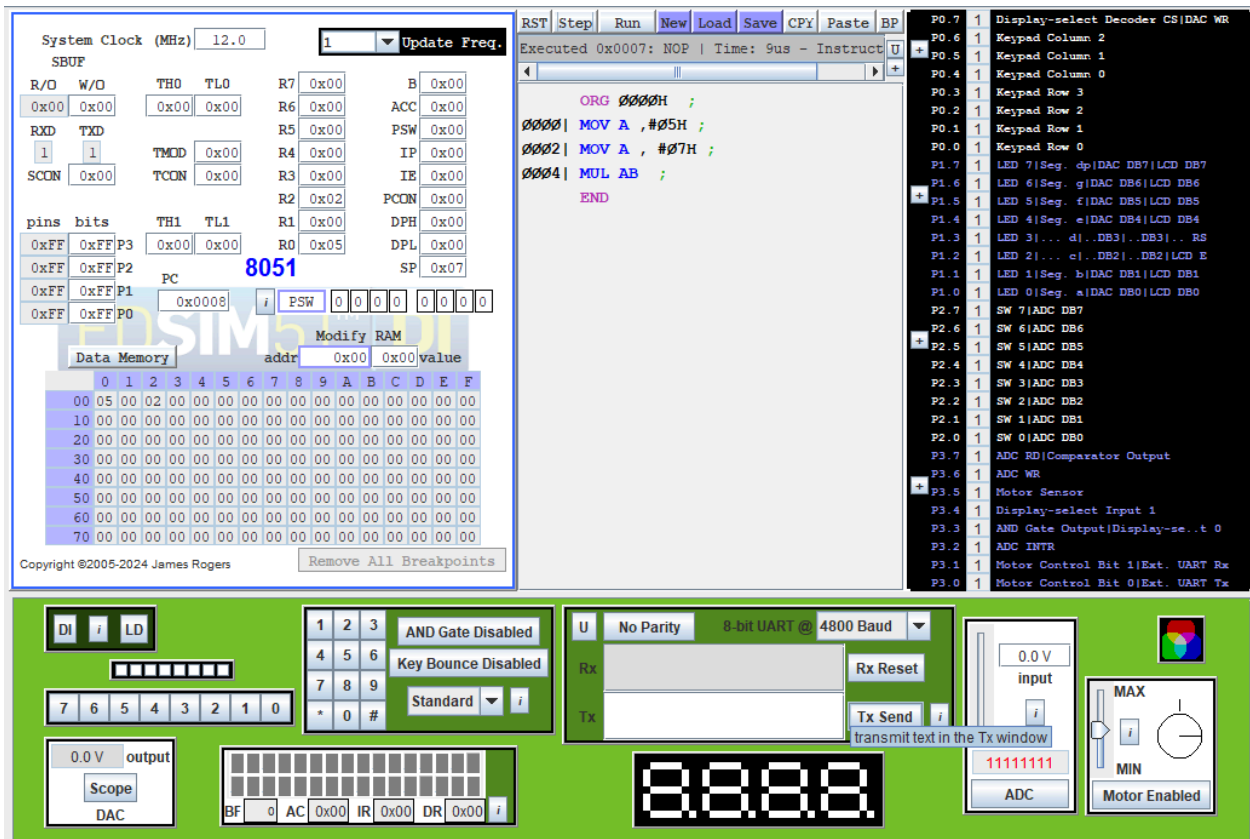
ORG 0000H ;

0000| MOV A, #05H ;

0002| MOV A, #07H ;

0004| MUL AB ;

END



(Q4) WRITE A PROGRAM TO DIVIDE 10H WITH 03H AND SAVE THE QUOTIENT IN R4 AND REMAINDER IN R5 ?

ORG 0000H ;

0000| MOV A, #10H ;

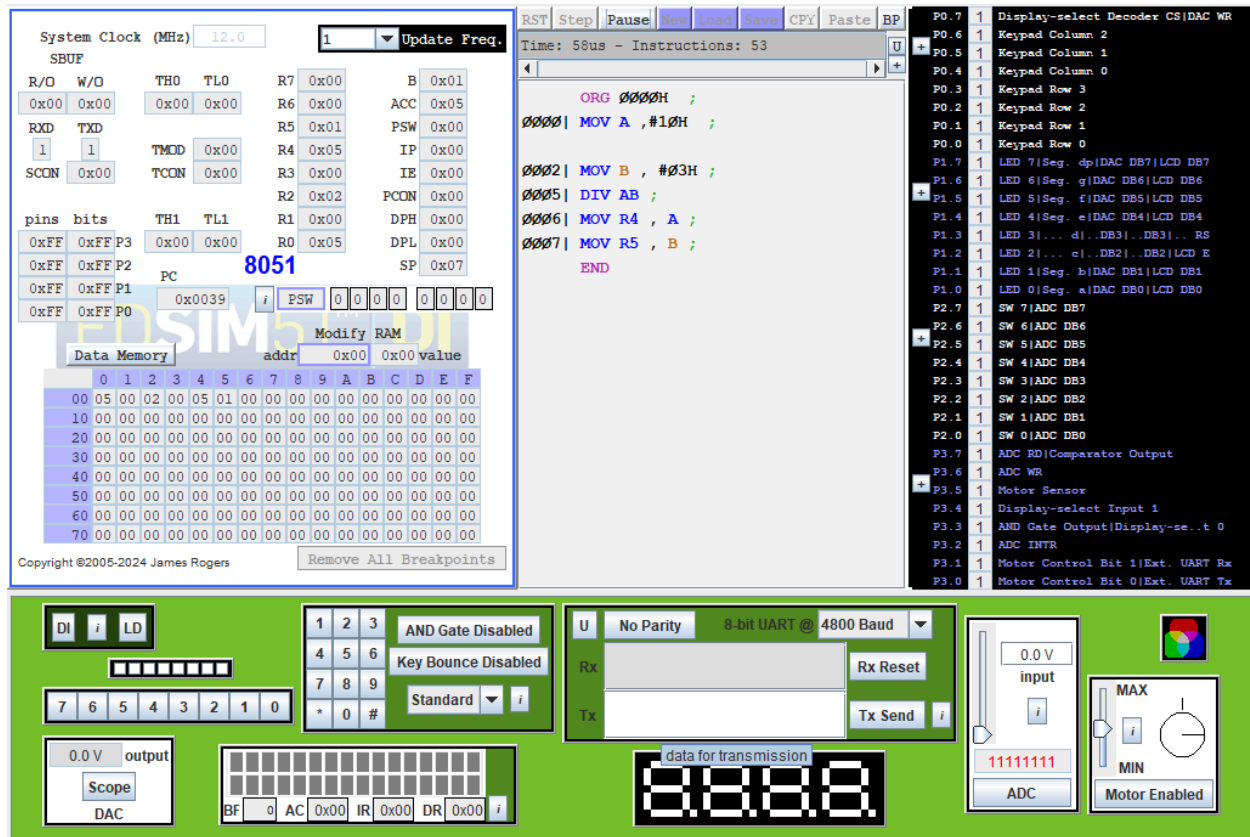
0002| MOV B, #03H ;

0005/ DIV AB ;

0006/ MOV R4 , A ;

0007/ MOV R5 , B ;

END



(Q5) WRITE A PORGRAM TO SWAP TWO NUMBERS 55H IS STORED IN R0 AND AAH IS STORED IN R1 . SWAP THE CONTENT OF R0 AND R1 ?

ORG 0000H ;

0000/ MOV R0,#55H ;

0002/ MOV R1 , #0AAH ;

0004/ MOV A , R0 ;

0005/ MOV R2,A ;

The screenshot displays the Proteus ISIS simulation environment. The central window shows the assembly code editor with the following code:

```
ORG 0000H ;
0000 MOV R0, #55H ;
0002 MOV R1, #0AAH ;
0004 MOV A, R0 ;
0005 MOV R2, A ;
0006 MOV A, R1 ;
0007 MOV R0, A ;
0008 MOV A, R2 ;
0009 MOV R1, A ;
END
```

The left panel shows the register and memory windows. The register window displays the following values:

Register	Value
R0	0x55
R1	0xAA
R2	0x55
R3	0x00
R4	0x05
R5	0x01
R6	0x00
R7	0x00
B	0x00
ACC	0x55
PSW	0x00
IP	0x00
IE	0x00
P0CON	0x00
DPL	0x00
DPH	0x00
SP	0x07

The memory window shows the following values:

Address	Value
0x00	0x55
0x01	0x55
0x02	0x55
0x03	0x55
0x04	0x55
0x05	0x55
0x06	0x55
0x07	0x55
0x08	0x55
0x09	0x55
0x0A	0x55
0x0B	0x55
0x0C	0x55
0x0D	0x55
0x0E	0x55
0x0F	0x55

The right panel shows the pin list for the microcontroller, including pins for the display-select decoder, keypad, and various I/O functions.

The bottom panel shows the physical circuit board, which includes a microcontroller, a display, a keypad, and various I/O components. The display shows the value 0.0 V, and the keypad shows the value 11111111.



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(Q6) WRITE A PROGRAM TO BLINK THE LED ON PORT 1 ?

ORG 0000H

START :

0000| MOV P1 , #00H ;

0003| MOV P1 , #0FFH ;

DELAY :

0006| MOV R6 , #255

0008| D1 : MOV R7 , #255

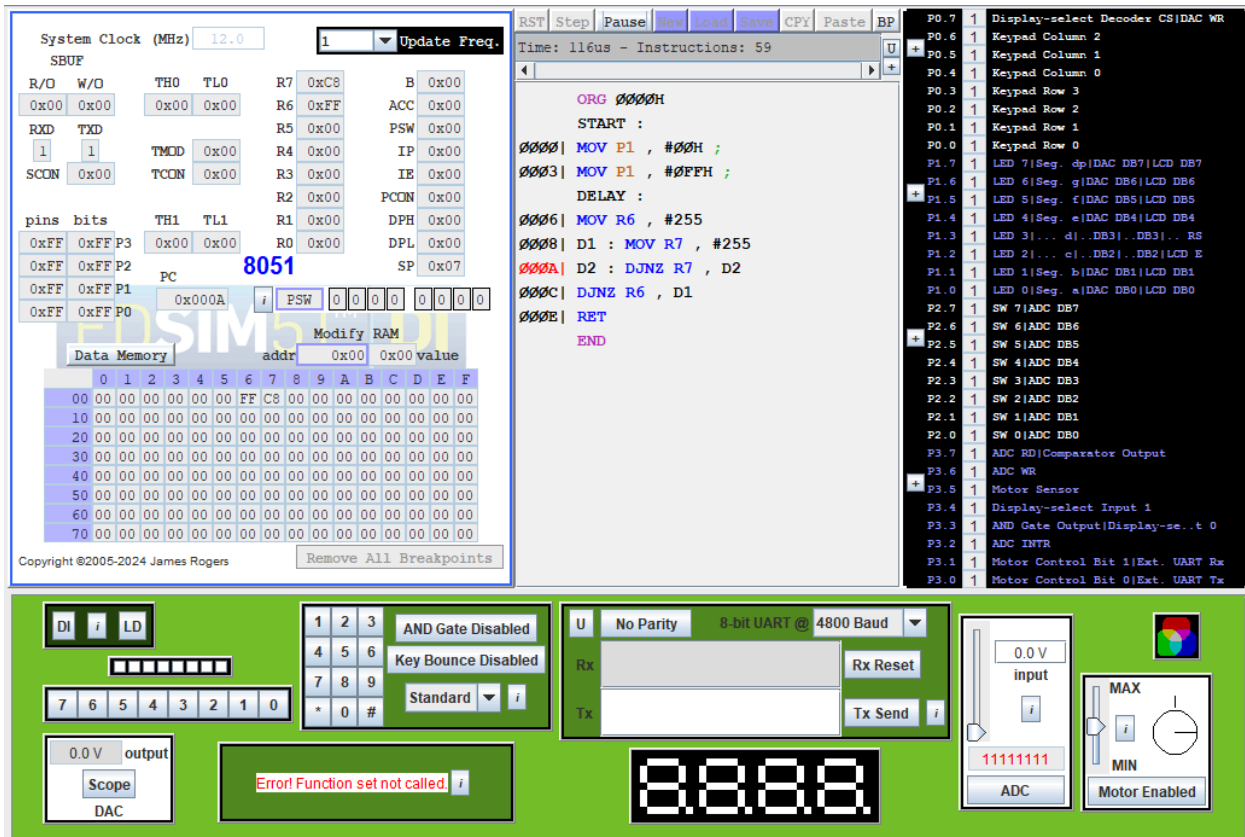
000A| D2 : DJNZ R7 , D2

000C| DJNZ R6 , D1

000E| RET

END





(7) write a program to blink the led from right to left ?

ORG 0000H

START:

MOV A,#01H

LOOP:

MOV P1,A

ACALL DELAY

RL A

JNZ LOOP

SJMP START

DELAY:

MOV R6,#255

D1: MOV R7,#255

D2: DJNZ R7,D2

DJNZ R6,D1

RET

END

The screenshot displays a microcontroller simulator interface with the following components:

- System Configuration:** System Clock (MHz) set to 12.0, Update Freq. button.
- Registers:** R0-R7, B, ACC, PSW, IP, PCON, DPH, DPL, SP. R6 is highlighted with value 0x00.
- Assembly Code:**

```
ORG 0000H
START:
0000 MOV A,#01H
LOOP:
0002 MOV P1,A
0004 ACALL DELAY
0006 RL A
0007 JNZ LOOP
0009 SJMP START
DELAY:
000B MOV R6,#255
000D D1: MOV R7,#255
000F D2: DJNZ R7,D2
0011 DJNZ R6,D1
0013 RET
```
- Data Memory:** Address 0x000F highlighted, value 0x00.
- Hardware Components:** Display-select Decoder CS|DAC WR, Keypad Column 2, Keypad Column 1, Keypad Column 0, Keypad Row 3, Keypad Row 2, Keypad Row 1, Keypad Row 0, LED 7|Seg. dp|DAC DB7|LCD DB7, LED 6|Seg. g|DAC DB6|LCD DB6, LED 5|Seg. f|DAC DB5|LCD DB5, LED 4|Seg. e|DAC DB4|LCD DB4, LED 3|... d|...DB3|...DB3|...RS, LED 2|... c|...DB2|...DB2|...LCD E, LED 1|Seg. b|DAC DB1|LCD DB1, LED 0|Seg. a|DAC DB0|LCD DB0, SW 7|ADC DB7, SW 6|ADC DB6, SW 5|ADC DB5, SW 4|ADC DB4, SW 3|ADC DB3, SW 2|ADC DB2, SW 1|ADC DB1, SW 0|ADC DB0, ADC RD|Comparator Output, ADC WR, Motor Sensor, Display-select Input 1, AND Gate Output|Display-se..t 0, ADC INTR, Motor Control Bit 1|Ext. UART Rx, Motor Control Bit 0|Ext. UART Tx.
- Hardware Controls:** DI, LD, AND Gate Disabled, Key Bounce Disabled, Standard, Rx, Tx, Rx Reset, Tx Send, 0.0 V input, 0.0 V output, Scope, DAC, Error! Function set not called, 11111111, ADC, MAX, MIN, Motor Enabled.

(8) WRITE A PROGRAM TO DISPLAY 8050 on a 7 segment display.

ORG 0000H

START:

MOV P1,#7FH ; 8

ACALL DELAY

MOV P1,#3FH ; 0

ACALL DELAY

MOV P1,#6DH ; 5

ACALL DELAY

MOV P1,#06H ; 1

ACALL DELAY

SJMP START

DELAY:

MOV R6,#255

D1: MOV R7,#255

D2: DJNZ R7,D2

DJNZ R6,D1

RET



END

The screenshot displays the Proteus 8.10 SP3 environment. The top section shows the assembly code editor for an 8051 microcontroller. The code includes initialization of registers, a delay loop, and a return statement. The hardware configuration window at the bottom shows various components like the 8051 microcontroller, 8-bit UART, and various peripheral modules.

Assembly Code:

```
MOV P1, #7FH ; 8
ACALL DELAY
MOV P1, #3FH ; 0
ACALL DELAY
MOV P1, #6DH ; 5
ACALL DELAY
MOV P1, #06H ; 1
ACALL DELAY
SJMP START

DELAY:
MOV R6, #255
D1: MOV R7, #255
D2: DJNZ R7, D2
DJNZ R6, D1
RET
```

Hardware Configuration:

- 8051 Microcontroller
- 8-bit UART @ 4800 Baud
- ADC (Analog-to-Digital Converter)
- Motor Control
- Display-select Decoder
- Keypad
- LEDs
- SW (Switches)
- ADC RD (Comparator Output)
- ADC WR
- Motor Sensor
- Display-select Input 1
- AND Gate Output (Display-select 0)
- ADC INTR
- Motor Control Bit 1 (Ext. UART Rx)
- Motor Control Bit 0 (Ext. UART Tx)

EXPERIMENT

TO BE PERFORMED SIMULATE & VALIDATE & VALIDATE
EXPERIMENTALLY

```
int ledPin = 13;
```

```
void setup() {  
  pinMode(ledPin, OUTPUT);  
}
```

```
void loop() {  
  digitalWrite(ledPin, HIGH); // Turn LED ON  
  delay(1000);               // Wait 1 second  
  
  digitalWrite(ledPin, LOW); // Turn LED OFF  
  delay(1000);               // Wait 1 second  
}
```



